Bangla Cyberbullying Detection Using Transformer Models and Ensemble Learning Techniques

Thesis submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Engineering

by

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DECLARATION

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We declare that this thesis entitled *Bangla Cyberbullying Detection Using Transformer Models and Ensemble Learning Techniques* is the result of my work except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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APPROVAL

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Dedicated to my parents
and respectable teachers

– *Rohed Deb Ove*

To my beloved parents the
source of my inspiration

– *Tamjidul Hasan*

Dedicated to my teachers
and inspiring mentors

– *Tahmim Ahmed*

ABSTRACT

Cyberbullying has emerged as a significant social concern on online platforms, particularly in low-resource languages such as Bangla. The informal nature of social media text, combined with linguistic variation and class imbalance, makes automated detection challenging. This study presents a comprehensive framework for Bangla cyberbullying detection using classical machine learning models, transformer-based architectures, and an ensemble approach. A multi-stage dataset construction process was followed, where an initial 44k category dataset was used to create an 18k binary dataset, later converted into a multi-class format and merged to produce a final dataset containing 60,153 labeled comments across five classes: Not Bully, Sexual, Religious, Threat, and Troll.

A two-stage classical model using TF-IDF features with Linear SVM and Logistic Regression was implemented as a baseline. Transformer-based models, including Bangla BERT and XLM-RoBERTa, were then fine-tuned for multi-class classification. To further enhance performance and reduce prediction variance, a soft-voting ensemble combining BERT and XLM-R was developed. The ensemble integrates probability outputs from both transformer models to improve robustness and stability across imbalanced categories.

Experimental results demonstrate that transformer models significantly outperform classical approaches. XLM-R achieves stronger macro-level performance than BERT, particularly for minority classes. The ensemble model delivers the best overall results, achieving 92.20% accuracy and 91.82% macro F1-score, indicating improved robustness and balanced class detection. These findings highlight the effectiveness of contextual transformer architectures and ensemble learning for Bangla cyberbullying detection and contribute toward developing reliable moderation systems for low-resource languages.

Keywords: Bangla Cyberbullying, NLP, BERT, XLM-R, Ensemble Learning, Multi-Class Classification, Social Media Analysis.

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Chapter 1

Introduction

1.1 Background of the Study

The rapid expansion of internet access and social media usage has significantly reshaped communication patterns across Bangladesh over the past decade. Platforms such as Facebook, YouTube, Twitter, and various online forums have become integral spaces for public interaction, opinion sharing, and social engagement. These digital environments allow individuals to connect beyond geographical limitations and actively participate in social discourse. However, alongside these advantages, the rise of online communication has also facilitated new forms of harmful behavior, particularly cyberbullying.

In Bangla-speaking online communities, cyberbullying often takes multiple forms rather than appearing as a single uniform behavior. Users may engage in trolling, religious hostility, sexually explicit harassment, or threatening language. These expressions are frequently shaped by socio-cultural dynamics such as religious sensitivity, gender bias, political affiliation, and social hierarchy. As a result, abusive content in Bangla digital spaces tends to be diverse and context-dependent.

Despite the growing prevalence of cyberbullying in Bangla online environments, most automated detection systems developed globally are designed for English language content. These systems typically rely on binary classification methods that categorize online text as either bullying or non-bullying. While this approach may be effective for basic identification, it fails to capture the nuanced nature of abuse present in Bangla discourse.

The limitation of binary detection is particularly evident in culturally complex environments such as Bangladesh, where different forms of online hostility carry varying

social implications. For instance, religious insults differ significantly from personal threats or trolling in terms of intent and impact.

The absence of category-based cyberbullying detection frameworks tailored for Bangla has therefore created a substantial gap in the field of natural language processing. Addressing this limitation requires a shift from simple detection toward classification of abuse types. In response to this need, the present study aims to develop a multi-class Bangla cyberbullying detection system capable of recognizing distinct forms of online harassment.

1.2 Problem Statement

Existing cyberbullying detection approaches in Bangla predominantly rely on binary classification techniques that label content as either bullying or non-bullying. Although such systems are useful for identifying harmful content, they provide limited insight into the nature of the abuse involved. In real-world online interactions, cyberbullying manifests in various forms including:

- Trolling
- Sexual harassment
- Religious hate speech
- Threatening language

Each of these forms represents a different level of severity and intent. However, when all abusive content is grouped under a single label, the effectiveness of automated moderation systems is reduced. For example, the response required for a threat differs significantly from that needed for trolling or religious hostility.

Another major challenge lies in dataset availability. Publicly accessible Bangla cyberbullying datasets are often small, imbalanced, or limited to binary annotations. This restricts the development of advanced classification models capable of understanding complex abusive patterns. Consequently, there exists a critical need to develop a comprehensive system that not only detects cyberbullying but also categorizes it into meaningful classes.

1.3 Research Objectives

The primary aim of this research is to design and implement a robust and scalable category-based cyberbullying detection framework tailored for Bangla language environments. Unlike conventional approaches that focus solely on identifying whether content is abusive or not, this study seeks to provide deeper insight into the nature and type of harmful interactions occurring in online Bangla communities.

To achieve this goal, the research is structured around the following objectives:

1. To construct a comprehensive and diversified Bangla cyberbullying dataset. This is achieved not only by utilizing existing publicly available datasets but also by transforming binary-labeled data into multi-class format and integrating multiple data sources. Through this transformation and merging process, the research seeks to overcome limitations associated with small or single-source datasets commonly used in previous studies.

2. To develop models capable of identifying distinct forms of cyberbullying. The study focuses on distinguishing categories such as trolling, sexual harassment, religious hostility, and threatening language, enabling more nuanced identification of harmful content.

3. To evaluate classical and transformer-based modeling approaches. A comparative evaluation is conducted across multiple modeling families to assess their effectiveness in capturing contextual patterns in Bangla cyberbullying detection.

4. To improve predictive performance using ensemble learning. By combining the strengths of multiple transformer models, the research aims to enhance classification accuracy and improve robustness in multi-class detection.

1.4 Motivation

The growing influence of digital communication has made social media platforms an essential part of everyday interaction in Bangladesh. While these platforms provide opportunities for social engagement and expression, they have also created an environment where harmful behavior such as cyberbullying can easily emerge and spread.

In Bangla-speaking online communities, cyberbullying is not limited to a single form. It frequently appears through trolling, sexual harassment, religious hostility, and threatening expressions. These forms of abuse often target sensitive social issues and can have serious psychological and emotional consequences for individuals.

Existing detection systems that rely on binary classification are insufficient to capture the complexity of such behaviors. Simply identifying whether content is abusive does not provide meaningful insight into the type of harm being caused. Different forms of cyberbullying require different levels of attention and intervention. For example, threatening language may demand immediate response, whereas trolling might require a different moderation strategy.

Moreover, the availability of large and well-annotated Bangla datasets remains limited, which restricts the development of advanced detection systems. Previous studies often rely on small datasets or focus only on identifying the presence of abuse rather than understanding its nature. This research is motivated by the need to move beyond basic detection toward a more comprehensive classification of cyberbullying. By identifying specific categories of abusive content, automated systems can become more effective tools for moderation, policy enforcement, and online safety.

Ultimately, the motivation of this study lies in contributing toward the development of intelligent systems capable of improving the quality of digital interactions in Bangla-speaking communities.

1.5 Significance of the Study

This research holds significant importance in advancing the field of Bangla Natural Language Processing, particularly in the domain of online safety and automated moderation.

One of the key contributions of this study lies in the development of a large-scale multi-class Bangla cyberbullying dataset. By transforming binary-labeled data into category-based annotations and merging datasets from different sources, the research addresses one of the major limitations of previous works—the reliance on small or single-source datasets.

Another important aspect of the study is the introduction of fine-grained classification of abusive language. Instead of merely identifying whether content is harmful,

the proposed system distinguishes between different types of cyberbullying. This enables a deeper understanding of online hostility and allows moderation systems to respond more effectively.

Furthermore, the research demonstrates the effectiveness of transformer-based ensemble models in improving classification performance for low-resource languages like Bangla. While previous studies have applied deep learning or transformer-based models individually, this work explores the benefits of combining multiple contextual models to enhance prediction accuracy.

From an application perspective, the proposed framework has the potential to support the development of intelligent moderation tools for social media platforms. By enabling more precise identification of harmful content, such systems can contribute to safer digital environments for Bangla-speaking users. In summary, this study not only advances academic research in Bangla NLP but also provides practical implications for improving online safety and content moderation.

1.6 Contributions and Thesis Organization

The key contributions of this thesis are summarized as follows:

- A systematic transformation framework that converts a binary Bangla cyberbullying dataset into a structured multi-class format, enabling fine-grained analysis of abuse categories such as *troll*, *sexual*, *religious*, and *threat*.

- The construction of a large-scale unified multi-class dataset (over 60,000 samples) by merging the transformed dataset with an existing category-annotated corpus, improving diversity and reducing the limitations of single-source data.

- A comprehensive experimental comparison of traditional machine learning baselines (e.g., TF–IDF with linear classifiers) against transformer-based models for multi-class Bangla cyberbullying detection.

- The fine-tuning and evaluation of a Bangla-focused BERT-based classifier to model contextual patterns in Bangla abusive text, serving as a strong transformer baseline in the proposed framework.

- The fine-tuning and evaluation of an XLM-R based classifier to leverage multilingual contextual representations for Bangla cyberbullying detection and to compare robustness across transformer architectures.
- An ensemble-based prediction strategy that combines the strengths of the BERT and XLM-R models (e.g., via soft voting / probability averaging) to improve classification accuracy and robustness over standalone models.
- The development of a real-time prediction interface that demonstrates the practical usability of the proposed ensemble model for cyberbullying detection and visualization.

The remainder of this thesis is organized as follows. Chapter 2 reviews the literature on cyberbullying detection and summarizes key findings and limitations in Bangla-focused studies. Chapter 3 presents the methodology, including dataset sources, dataset transformation and merging procedures, preprocessing steps, and model configurations. Chapter 4 describes the design and development of the proposed system, detailing classical baselines, transformer fine-tuning (BERT and XLM-R), and the ensemble integration strategy. Chapter 5 reports the experimental results and analysis, including performance comparisons, confusion matrices, and error analysis with a focus on the ensemble model. Finally, Chapter 6 concludes the thesis by summarizing the findings, discussing limitations, and outlining directions for future work.

Chapter 2

Literature Review

2.1 Related Work

Cyberbullying detection has become a crucial research area in Natural Language Processing (NLP) as social media platforms continue to grow rapidly. Automated systems capable of detecting abusive and harmful language are essential for maintaining safe online environments. Early approaches to cyberbullying detection relied primarily on rule-based methods and keyword filtering techniques. These systems used predefined dictionaries of offensive terms and pattern-matching rules to detect abusive comments. Although these methods are computationally efficient and easy to interpret, they often fail to capture contextual meaning and implicit forms of harassment.

With the advancement of machine learning, researchers began applying statistical classifiers for text classification tasks. Feature extraction techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) became widely used to represent textual data numerically [1]. Using such representations, classifiers like Support Vector Machines (SVM) have demonstrated strong performance in high-dimensional text classification problems [2]. Efficient implementations such as LIBLINEAR further improved scalability for large-scale linear classification tasks [3]. These classical machine learning approaches provide strong baseline performance but rely heavily on manual feature engineering and cannot fully capture contextual semantics.

Before the rise of deep contextual models, several studies explored distributed word representations such as Word2Vec to capture semantic relationships between words in continuous vector spaces [4]. Efficient text classification models such as FastText were later proposed to further improve training efficiency and scalability for

large text corpora [5]. These techniques significantly improved the representation of textual data for downstream machine learning models.

The introduction of neural networks significantly improved the ability of models to learn complex patterns from textual data. Neural networks trained through backpropagation allow models to automatically learn hierarchical representations of input features [6]. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) architectures were widely applied to sequence modeling tasks, enabling systems to capture contextual dependencies between words [7]. Despite these improvements, recurrent models often suffer from limitations in capturing long-range dependencies and require extensive computational resources.

A major breakthrough in NLP came with the introduction of the Transformer architecture, which uses self-attention mechanisms to model relationships between words in a sentence [8]. Transformer-based models allow parallel computation and significantly improve contextual understanding compared to earlier architectures.

Building upon this architecture, Bidirectional Encoder Representations from Transformers (BERT) introduced bidirectional contextual embeddings trained on large-scale corpora [9]. BERT achieved state-of-the-art performance across many NLP tasks including sentiment analysis, question answering, and text classification.

Another powerful multilingual model is XLM-RoBERTa (XLM-R), which learns cross-lingual representations across multiple languages using large multilingual datasets [10]. Multilingual transformer models are particularly beneficial for low-resource languages because they enable knowledge transfer across languages.

The development of open-source NLP libraries has further accelerated research progress. The HuggingFace Transformers framework provides efficient tools for implementing and fine-tuning pretrained transformer models [11]. These frameworks have enabled researchers to easily apply transformer architectures to various NLP tasks.

Beyond individual models, ensemble learning techniques have been widely used to improve classification performance. Ensemble methods combine predictions from multiple models to reduce variance and improve predictive accuracy [12]. Combining multiple classifiers often produces more stable predictions than relying on a single model [13].

Evaluation metrics also play an important role in NLP systems. For text classification tasks, metrics such as accuracy, precision, recall, and F1-score are commonly

used to measure model performance [14]. In language generation and translation tasks, metrics such as BLEU and SacreBLEU are widely used for automatic evaluation [15, 16].

In recent years, research has also considered broader societal impacts of NLP models. Studies have highlighted environmental costs associated with large-scale neural models [17] and ethical concerns regarding large language models [18]. Additionally, frameworks such as model cards and datasheets for datasets have been proposed to improve transparency and responsible AI practices [19, 20].

Overall, existing literature shows that cyberbullying detection has evolved from rule-based filtering methods to advanced transformer-based architectures. Classical machine learning models provide strong baseline performance, while transformer models and ensemble approaches offer improved contextual understanding and classification accuracy.

2.2 Research Gap

Despite the rapid progress in cyberbullying detection research, several challenges remain, particularly for low-resource languages such as Bangla.

First, most existing research focuses primarily on high-resource languages such as English, where large annotated datasets are readily available. In contrast, Bangla cyberbullying detection research remains limited due to the scarcity of large-scale labeled datasets and the lack of standardized benchmark corpora.

Second, many previous studies concentrate mainly on binary classification tasks, where comments are categorized simply as bullying or non-bullying. Although binary classification is useful for basic moderation systems, it fails to capture the complexity of abusive language. Multi-class categorization that distinguishes between different forms of cyberbullying such as sexual harassment, religious abuse, trolling, and threats remains relatively underexplored.

Third, existing studies often rely on a single modeling approach. While classical machine learning models provide strong baselines and transformer models achieve high contextual accuracy, few studies conduct comprehensive comparisons across classical models, transformer architectures, and ensemble techniques within a unified experimental framework.

Another limitation lies in dataset construction methodologies. Many existing datasets are collected from single sources and directly used for model training. Limited research has explored systematic transformation of binary cyberbullying datasets into category-based datasets and merging them with existing resources to construct larger and more diverse corpora.

Furthermore, Bangla social media text often contains informal language, spelling variations, and mixed linguistic patterns, which make cyberbullying detection more challenging. Existing studies rarely address these complexities when building datasets or designing classification models.

Finally, although transformer models such as BERT and XLM-R demonstrate strong performance individually, ensemble approaches combining multiple contextual models remain relatively underexplored in Bangla cyberbullying detection. Leveraging the complementary strengths of multiple transformer architectures may significantly improve classification robustness and overall performance.

To address these limitations, this research constructs a large-scale multi-class dataset containing over 60,000 labeled comments by transforming and merging multiple datasets. The study further evaluates classical machine learning models, transformer-based architectures (BERT and XLM-R), and a soft-voting ensemble approach to identify the most effective framework for Bangla cyberbullying detection.

Chapter 3

Methodology

3.1 Dataset Description

The dataset used in this research was constructed through a multi-stage transformation process to ensure label consistency, scalability, and experimental reliability. Rather than directly training on a single raw dataset, a structured pipeline was followed: starting from an existing 44k category dataset, an 18k binary dataset was manually constructed, later converted back into a multi-category format, and finally merged to produce a 60,153-sample final category dataset. This systematic dataset engineering approach improves class representation and supports both binary and multi-class experiments.

3.1.1 44k Category Dataset

The 44k category dataset served as the foundational dataset for this research. It contains approximately 44,000 Bangla social media comments annotated into multiple cyberbullying categories. Each comment is labeled according to specific types of harmful behavior such as sexual harassment, religious attacks, threats, trolling, or non-bullying content.

The dataset reflects real-world linguistic characteristics of Bangla social media communication, including:

- Informal spelling variations
- Slang expressions
- Dialectal influences

- Code-mixed Bangla–English text
- Social-media specific abbreviations

This dataset was primarily used as a reference source for category mapping and label alignment during subsequent dataset construction stages.

3.1.2 18k Binary Dataset

To establish a baseline classification framework, an 18,000-sample binary dataset was constructed from the available labeled data. In this stage, all bullying-related categories were merged into a single positive class labeled "Bully", while non-harmful comments were labeled as "Not Bully."

The binary label structure was defined as:

- Not Bully $\rightarrow 0$
- Bully $\rightarrow 1$

The purpose of creating this dataset was to:

- Simplify the classification problem
- Evaluate baseline model performance
- Analyze model behavior under a reduced label space

The dataset was carefully balanced to reduce extreme class dominance and ensure fair evaluation.

3.1.3 18k Category Dataset

After constructing the binary dataset, the next step involved reconstructing multi-class labels. Using the original 44k dataset as a reference, the 18k binary samples were mapped back to their specific bullying categories.

- Each "Bully" instance was reassigned its original category label
- "Not Bully" instances remained unchanged
- Label alignment was verified through code-based mapping

This resulted in an 18k category dataset with consistent class labels:

- Not Bully
- Sexual
- Religious
- Threat
- Troll

This conversion allowed direct comparison between binary and multi-class classification tasks using a shared data foundation.

3.1.4 Final Merged Dataset

In the final stage, the 18k converted category dataset was merged with the original 44k category dataset to create the final training corpus.

After duplicate removal and label consistency verification, the merged dataset contained:

- 60,153 samples with 3 columns (text, label_text, encoded label)

The final label mapping used throughout the experiments was:

$$\{\text{not bully} : 0,\ \text{threat} : 1,\ \text{religious} : 2,\ \text{troll} : 3,\ \text{sexual} : 4\}.$$

The class distribution is moderately imbalanced, with "Not Bully" being the largest category and "Threat" being the smallest. This imbalance influenced later modeling decisions, such as the use of class-weighted loss in transformer training and macro F1-score evaluation.

The final 60k dataset serves as the primary dataset for all classical, transformer-based, and ensemble experiments presented in this thesis.

3.2 Data Preprocessing

Data preprocessing was performed to ensure data consistency, reduce noise, and prepare the dataset for both classical machine learning and transformer-based models. Since the dataset consists of Bangla social media comments, careful preprocessing was required to preserve semantic meaning while eliminating irrelevant patterns.

3.2.1 Cleaning

The cleaning stage focused on standardizing textual input while removing unnecessary noise. The following steps were applied:

- Lowercasing: All comments were converted to lowercase to maintain uniform vocabulary representation.
- Special Character Removal: Irrelevant symbols, excessive punctuation, and unsupported characters were removed while preserving meaningful textual content.
- Whitespace Normalization: Multiple spaces were replaced with single spaces to ensure consistent formatting.
- Duplicate Removal: Duplicate entries were removed prior to splitting to prevent data leakage and inflated performance.
- Null Handling: Empty or missing text entries were excluded.

For transformer-based experiments (BERT and XLM-R), cleaning was kept minimal to preserve contextual information. For classical models, cleaning ensured stable TF-IDF feature extraction.

3.2.2 Tokenization

Tokenization converts raw text into smaller processing units.

Classical Machine Learning Models. For classical approaches, word-level tokenization was applied. Text was split into whitespace-separated tokens before TF-IDF vectorization. Unigram and bigram features were extracted to capture local word patterns.

The TF-IDF representation is defined as:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log\left(\frac{N}{\text{DF}(t)}\right), \tag{3.1}$$

where $\text{TF}(t,d)$ is term frequency, $\text{DF}(t)$ is document frequency, and N is the total number of documents.

Transformer-Based Models. For BERT and XLM-R, pretrained tokenizers were used. These tokenizers apply subword segmentation techniques (WordPiece or SentencePiece), which:

- Handle rare words effectively
- Improve representation of morphologically rich Bangla text
- Support multilingual input

Sequences were padded or truncated to a fixed maximum length for stable batch training.

3.2.3 Label Encoding

Since machine learning models require numerical labels, categorical labels were converted into integer format using a consistent mapping:

- Not Bully $\rightarrow 0$
- Threat $\rightarrow 1$
- Religious $\rightarrow 2$
- Troll $\rightarrow 3$
- Sexual $\rightarrow 4$

This encoding ensures compatibility with the cross-entropy loss function:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \tag{3.2}$$

where C represents the total number of classes.

The label mapping was preserved during training and inference to maintain consistency across models and ensemble evaluation.

3.2.4 Train-Test Split

To ensure fair performance evaluation, the dataset was divided into separate subsets.

Classical Machine Learning.

- 80% Training
- 20% Testing

Stratified sampling was applied to preserve class distribution.

Transformer-Based Models. A three-way split was applied using group-based splitting to prevent data leakage:

- 72% Training
- 8% Validation
- 20% Testing

The validation set was used for hyperparameter tuning and early stopping, while the test set remained unseen until final evaluation.

3.3 Dataset Statistics

After merging the 44k category dataset with the 18k converted category dataset and removing duplicate entries, the final dataset consists of:

60,153 samples and 3 columns

text	label_text	label
দেশ থেকে বিতাড়িত করা হোক	threat	1
আপনি আমাদের তারন্যের অহংকার লিডার	not bully	0
কেন ফাক? আলোচ্য বিষয়টি কি?	not bully	0
অর্থ সহ কোরআন পড়েই আজ আমি নাস্তিক।	religious	2
সে এমন অভদ্রতা করছে কেন? বুঝলাম না তো ।	not bully	0

Table 3.1: Sample Dataset Preview

The three columns include:

- Text (Bangla comment)
- label_text (categorical label)
- Encoded label (numerical representation)

This dataset was used for all experiments involving classical models, transformer-based models, and ensemble learning.

3.3.1 Class Distribution

The final dataset contains five classes. The label encoding scheme is:

$$\{\text{not bully} : 0,\ \text{threat} : 1,\ \text{religious} : 2,\ \text{troll} : 3,\ \text{sexual} : 4\}.$$

The distribution of samples across classes is shown below.

Class Name	Number of Samples	Percentage (%)
Not Bully	26,604	44.23%
Troll	13,522	22.47%
Sexual	10,558	17.55%
Religious	7,691	12.79%
Threat	1,778	2.96%
Total	60,153	100%

Table 3.2: Class Distribution of Final Dataset

From the distribution, it is evident that:

- The dataset is moderately imbalanced.
- The "Threat" category is significantly underrepresented.
- "Not Bully" constitutes the largest portion of the dataset.

This imbalance influenced later modeling decisions, including the use of class balancing in classical models and weighted loss functions in transformer-based training.

3.3.2 Data Split Overview

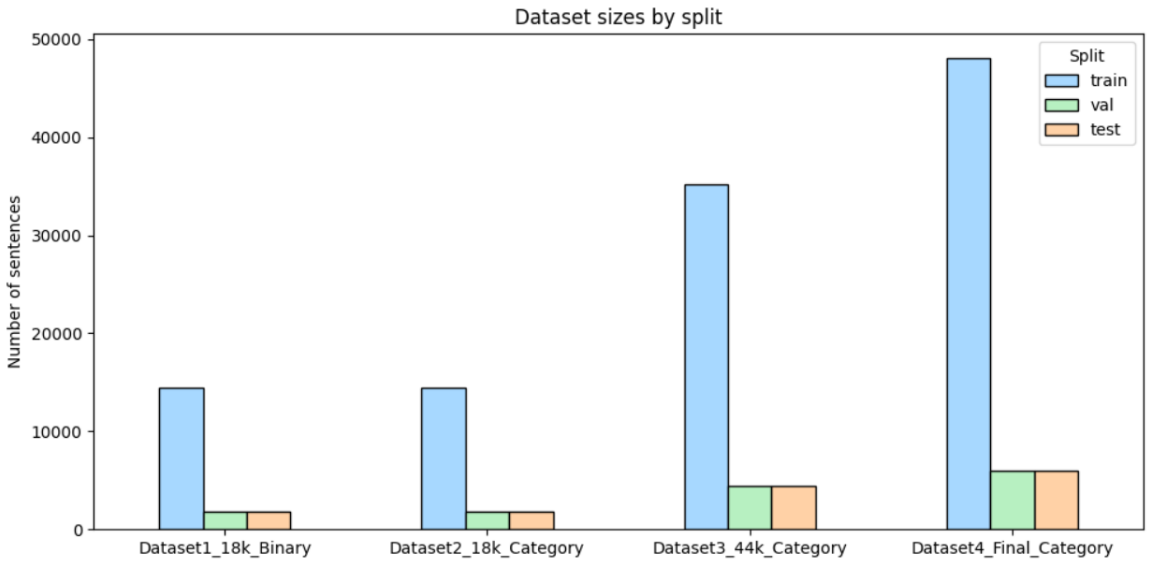

Figure 3.1: Number of Sentences Across Train, Validation and Test Splits

Figure 3.1 shows the distribution of sentences across training, validation, and test splits for all constructed datasets. The Final Category dataset contains the highest number of samples, followed by the 44k Category dataset, while the 18k datasets are comparatively smaller.

In each case, most data is allocated to the training split, with smaller portions reserved for validation and testing. This consistent split strategy ensures fair evaluation and supports reliable comparison across different modeling approaches.

The consistent split ratios ensure balanced evaluation and reduce potential sampling bias.

3.3.3 Dataset Statistics Overview

Dataset	Split	Sentences	Avg Tokens	P50	P90	P95
18k Binary	Train	14,400	16.02	13	27	35.05
	Validation	1,800	16.27	13	27.1	36.00
	Test	1,800	17.26	13	29	39.00
18k Category	Train	14,400	16.21	13	27	36.00
	Validation	1,800	15.81	13	27	35.00
	Test	1,800	15.66	12	26.1	33.00
44k Category	Train	35,200	16.67	9	38	59.00
	Validation	4,400	16.74	9	37	58.00
	Test	4,401	16.04	9	36	55.00
Final Category	Train	48,122	16.55	10	34	52.00
	Validation	6,015	16.60	10	33	52.00
	Test	6,016	16.74	10	34	51.00

Table 3.3: Dataset Statistics After Preprocessing

Table 3.3.3 presents the dataset statistics after preprocessing, including the number of sentences and token-length summaries across training, validation, and test splits for each constructed dataset. These statistics provide insight into dataset scale, distribution, and textual characteristics.

The Final Category dataset is the largest corpus, containing 48,122 training sentences, making it suitable for transformer fine-tuning. The 44k Category dataset also provides substantial training data, while the 18k Binary and 18k Category datasets serve as intermediate and baseline evaluation corpora.

3.3.4 Sentence Length Characteristics

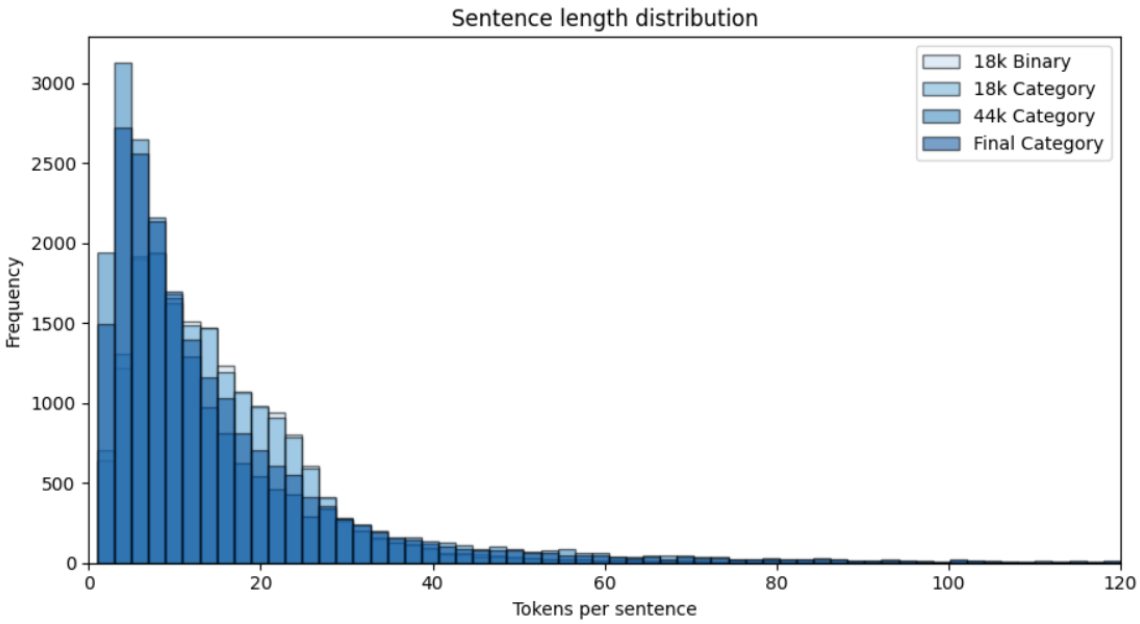

Figure 3.2: Distribution of Sentence Lengths Across Datasets

Figure 3.2 illustrates the distribution of sentence lengths (measured in tokens) across the four constructed datasets: 18k Binary, 18k Category, 44k Category, and Final Category. For clarity and readability, the distribution is clipped at 120 tokens.

The histogram reveals important characteristics of Bangla social media text:

- **Predominantly Short Sentences:** Most comments fall within the 5–20 token range. This reflects the informal and concise nature of user-generated social media content.
- **Right-Skewed Distribution:** All datasets exhibit a right-skewed distribution, where shorter sentences dominate and longer comments gradually decrease in frequency.
- **Moderate Tail Behavior:** Although the majority of sentences are short, a small number of longer comments extend beyond 60 tokens. The 44k and Final Category datasets show slightly heavier tails compared to the 18k datasets, indicating greater variability in sentence structure.

- **Consistency Across Datasets:** The overall length distribution remains relatively consistent between the 18k and 44k datasets, suggesting that the merging process preserved natural linguistic characteristics.

3.4 Models

To evaluate Bangla cyberbullying detection performance, this research implemented four modeling approaches: classical machine learning models, deep learning models, transformer-based architectures, and an ensemble framework. These models were selected to compare traditional feature-based classification with contextual language modeling techniques.

3.4.1 Classical Machine Learning

The classical modeling framework was implemented using a two-stage classification pipeline.

Feature Representation. Textual features were extracted using TF-IDF vectorization with unigram and bigram representations. The TF-IDF weighting scheme is defined as:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log\left(\frac{N}{\text{DF}(t)}\right), \tag{3.3}$$

where $\text{TF}(t,d)$ is the frequency of term t in document d, $\text{DF}(t)$ is the document frequency of term t, and N is the total number of documents.

Stage-1: Binary Classification. The first stage distinguishes between:

- Not Bully
- Bully

A Linear Support Vector Machine (LinearSVC) was used with class balancing to handle label imbalance. Since LinearSVC does not directly produce probability outputs, probability calibration was applied using sigmoid calibration.

Stage-2: Multi-Class Classification. For samples predicted as "Bully" in Stage-1, a second classifier was applied to categorize them into:

- Sexual
- Religious
- Threat
- Troll

A Logistic Regression classifier with class balancing used for this multi-class stage.

This two-stage architecture reduces misclassification between non-bullying and specific bullying types.

3.4.2 Deep Learning Approaches

Deep learning models automatically learn hierarchical feature representations from raw text data without manual feature engineering.

Unlike classical methods, deep learning models process token embeddings and optimize parameters through back propagation using cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{i} y_i \log(\hat{y}_i), \tag{3.4}$$

where C is the number of classes, y_i is the true label indicator, and $\hat{y}_i$ is the predicted probability.

These models are capable of capturing contextual semantics and word dependencies, making them more suitable for complex linguistic patterns present in Bangla social media text.

3.4.3 Transformer-Based Models

Transformer architectures represent state-of-the-art NLP models due to their self-attention mechanism, which enables contextual representation of words.

The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \tag{3.5}$$

Two transformer-based models were fine-tuned in this research:

BERT: A pretrained Bangla BERT model was fine-tuned for multi-class cyberbullying detection.

Key characteristics:

- Bidirectional contextual embeddings
- Classification head added on top of the transformer encoder
- Maximum sequence length constraint
- Early stopping based on validation performance

The dataset was split into 72% training, 8% validation, and 20% testing. Training was performed using the Hugging Face Trainer API.

XLM-RoBERTa: XLM-RoBERTa is a multilingual transformer pretrained on large-scale multilingual corpora. It was fine-tuned for multi-class classification using the same train-validation-test split.

Distinct features of this implementation include:

- Larger contextual capacity compared to base BERT
- Class-weighted cross-entropy loss to address dataset imbalance
- Evaluation using macro F1-score to account for minority classes

XLM-R demonstrated improved robustness for underrepresented categories such as "Threat."

3.4.4 Ensemble Learning

To improve overall classification performance, a soft-voting ensemble was implemented by combining BERT and XLM-R probability outputs.

Instead of selecting predictions from a single model, ensemble probabilities were computed as:

$$\mathbf{P}_{\text{ens}} = w_1 \mathbf{P}_{\text{BERT}} + w_2 \mathbf{P}_{\text{XLM-R}} \tag{3.6}$$

where $w_1 + w_2 = 1$, $\mathbf{P}_{\text{BERT}}$ is the probability vector from BERT, and $\mathbf{P}_{\text{XLM-R}}$ is the probability vector from XLM-R.

The final predicted class is determined by:

$$\hat{y} = \arg\max\left(\mathbf{P}_{\text{ens}}\right) \tag{3.7}$$

Weight selection was performed experimentally to maximize validation accuracy and macro F1-score.

The ensemble approach leverages complementary strengths of both transformer models, leading to improved classification stability and better performance on minority classes.

3.5 Hyper parameter Tuning

Hyperparameter tuning plays a critical role in optimizing model performance and ensuring stable convergence during training. In this research, hyperparameters were selected empirically based on validation performance, computational constraints, and dataset characteristics.

Different tuning strategies were applied for classical machine learning models, transformer-based models, and the ensemble framework.

3.5.1 Classical Model Hyperparameters

For classical machine learning models, tuning focused on feature extraction and regularization parameters.

TF-IDF Parameters.

- N-gram range: (1, 2)
- Maximum features: controlled to reduce dimensionality
- Sublinear term frequency scaling

Linear SVM (Binary Stage).

- Class weight: balanced
- Regularization parameter C: tuned experimentally

Logistic Regression (Multi-Class Stage).

- Solver: liblinear
- Maximum iterations: increased to ensure convergence
- Class weight: balanced

Stratified train-test splitting ensured fair evaluation without the need for a separate validation set.

3.5.2 BERT Hyperparameters

For BERT fine-tuning, several hyperparameters were adjusted to achieve optimal validation performance.

Key parameters include:

- Learning rate: 1.5×10^{-5}
- Batch size: 8
- Gradient accumulation steps: 2
- Number of epochs: 4
- Maximum sequence length: 128
- Weight decay: small regularization value

Early stopping was applied based on validation loss to prevent overfitting. The best-performing model checkpoint was saved automatically.

3.5.3 XLM-R Hyperparameters

For XLM-RoBERTa, slightly different hyperparameters were used due to model size and training dynamics.

- Learning rate: 2×10^{-5}
- Batch size: 16
- Number of epochs: 6

- Maximum sequence length: 256
- Weight decay: 0.01
- Class-weighted loss: applied to address imbalance

Macro F1-score was used as the primary metric for model selection due to class imbalance.

3.5.4 Ensemble Weight Optimization

For the ensemble model, weight tuning was performed by searching over different combinations of probability weights assigned to BERT and XLM-R.

The ensemble probability is defined as:

$$\mathbf{P}_{\text{ens}} = w_1 \mathbf{P}_{\text{BERT}} + w_2 \mathbf{P}_{\text{XLM-R}} \tag{3.8}$$

where $w_1 + w_2 = 1$.

A grid search over weight values was conducted to maximize validation accuracy and macro F1-score. The optimal weight combination was selected based on overall performance stability.

Chapter 4

Design and Development

4.1 System Architecture

The proposed system follows a structured pipeline to detect Bangla cyberbullying using classical, transformer-based, and ensemble models. The architecture is modular to ensure consistency across experiments and fair performance comparison.

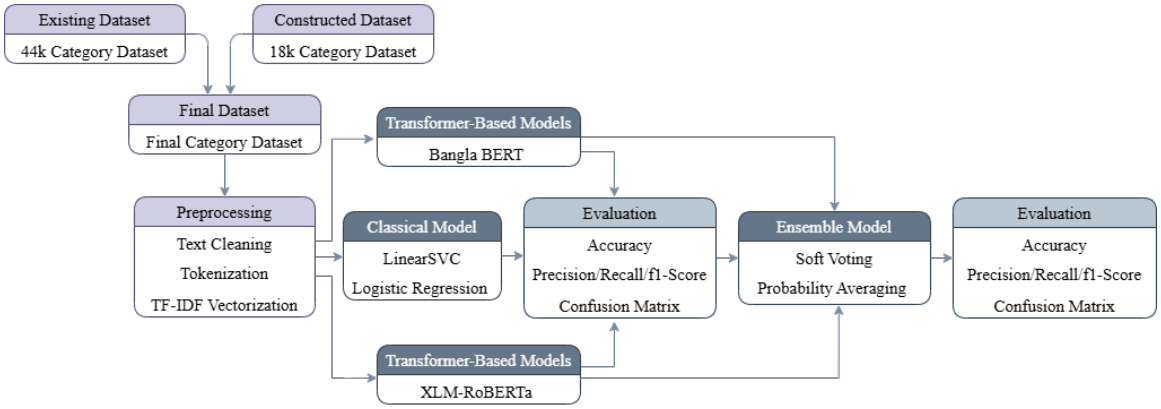

Figure 4.1: System Architecture

Workflow Overview

- **Dataset Input:** The final merged dataset (60,153 samples) is loaded with text and encoded labels.
- **Preprocessing Module:** Text cleaning, normalization, and tokenization are applied.

  - Classical models use TF-IDF vectorization.
  - BERT and XLM-R use pretrained tokenizers.

- **Data Splitting:**
  - Classical: 80% training, 20% testing
  - Transformers & Ensemble: 72% training, 8% validation, 20% testing

- **Model Training:**
  - Classical: Two-stage pipeline (Binary → Category)
  - BERT: Fine-tuned multi-class classifier
  - XLM-R: Fine-tuned with class-weighted loss

- **Ensemble Module:** BERT and XLM-R probabilities are combined using soft voting to improve classification robustness.

- **Evaluation Module:** Performance is measured using accuracy, precision, recall, F1-score, confusion matrix, and training curves.

4.2 Classical Model

The classical modeling framework was implemented as a two-stage classification pipeline. Instead of directly performing multi-class classification, the system first distinguishes between bullying and non-bullying content, and then classifies bullying comments into specific categories. This hierarchical approach improves classification stability and reduces confusion between unrelated classes.

4.2.1 Feature Extraction (TF-IDF)

For classical modeling, textual comments were transformed into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) method.

$$\text{TF-IDF}(t,d) = \text{tf}(t,d) \times \log\left(\frac{N}{\text{df}(t)}\right) \tag{4.1}$$

where $\text{TF}(t,d)$ is the frequency of term t in document d, $\text{DF}(t)$ is the number of documents containing term t, and N is the total number of documents.

In this implementation:

- Word-level tokenization was applied.
- Unigram and bigram features were extracted.
- TF-IDF vectors were generated separately for binary and multi-class stages.
- Sparse matrix representations were used for computational efficiency.

TF-IDF was chosen because it is effective for linear text classification and provides a strong baseline for comparison with deep learning models.

4.2.2 Logistic Regression & SVM

Two linear classifiers were used in the classical pipeline.

Stage-1: Binary Classification. The first stage classifies comments into:

- Not Bully
- Bully

A Linear Support Vector Machine was used due to its strong performance in high-dimensional sparse text spaces.

The SVM optimization objective is:

$$\min_{\mathbf{w}} \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i \tag{4.2}$$

where $\mathbf{w}$ represents the decision boundary, C controls regularization strength, and ξ_i represents classification error.

Class balancing was applied to address dataset imbalance. Since LinearSVC does not produce probability outputs directly, probability calibration was performed to enable probabilistic predictions when needed.

Stage-2: Multi-Class Classification. For comments predicted as "Bully" in Stage-1, a Logistic Regression classifier was applied to categorize them into:

- Sexual
- Religious
- Threat
- Troll

$$P(y = i \mid x) = \frac{e^{w_i^T x}}{\sum_{j=1}^{C} e^{w_j^T x}} \tag{4.3}$$

Class weighting was applied to handle imbalance among bullying categories.

4.2.3 Training Procedure

The classical model training followed these steps:

- Load and preprocess dataset
- Perform 80% training and 20% testing split (stratified)
- Fit TF-IDF vectorizer on training data
- Train Stage-1 binary SVM classifier
- Identify predicted bullying samples
- Train Stage-2 Logistic Regression classifier on bullying subset
- Evaluate performance using:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion Matrix

The two-stage architecture reduces false positives and improves category-level discrimination compared to single-step multi-class classification.

4.3 BERT Model

To capture contextual semantics in Bangla social media text, a pretrained Bangla BERT model was fine-tuned for multi-class cyberbullying detection. Unlike classical models that rely on manually engineered features, BERT generates contextual word embeddings using a transformer-based architecture with self-attention mechanisms.

4.3.1 Model Architecture

BERT (Bidirectional Encoder Representations from Transformers) is based on the transformer encoder architecture. It processes text bidirectionally, meaning each token representation is influenced by both left and right context.

The core component of BERT is the self-attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{4.4}$$

where Q is the Query matrix, K is the Key matrix, V is the Value matrix, and d_k is the scaling factor.

- A pretrained Bangla BERT base model was used.
- A classification head (fully connected layer) was added on top of the [CLS] token representation.
- The output layer dimension corresponds to 5 classes.
- The final prediction is obtained through a softmax activation over the class logits.

4.3.2 Fine-Tuning Strategy

Fine-tuning was performed using supervised learning on the 60k multi-class dataset.

- Tokenization using the pretrained BERT tokenizer.
- Maximum sequence length constraint (128 tokens).
- Padding and truncation applied to ensure uniform input length.

Dataset split:

- 72% Training
- 8% Validation
- 20% Testing

Training was conducted using the Hugging Face Trainer API with:

- Learning rate: 1.5×10^{-5}
- Batch size: 8
- Gradient accumulation: 2 steps
- Number of epochs: 4

Early stopping was applied based on validation performance to prevent overfitting. The best model checkpoint was saved automatically.

4.3.3 Loss Function

For multi-class classification, the Cross-Entropy Loss function was used:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{4.5}$$

where C is the number of classes, y_i is the true label indicator, and $\hat{y}_i$ is the predicted probability.

4.4 XLM-R Model

To enhance multilingual contextual understanding and improve performance on minority classes, XLM-RoBERTa (XLM-R) was fine-tuned for Bangla cyberbullying detection. XLM-R is a transformer-based multilingual model pretrained on large-scale cross-lingual corpora, making it suitable for morphologically rich and low-resource languages such as Bangla.

4.4.1 Model Architecture

XLM-R is built upon the transformer encoder architecture and utilizes a self-attention mechanism to generate contextual token representations.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{4.6}$$

In this implementation:

- A pretrained `xlm-roberta-base` model was used.
- A classification head was added on top of the pooled representation.
- The output layer dimension corresponds to 5 classes.
- Maximum sequence length was set to 256 tokens to capture longer contextual patterns.

Compared to BERT, XLM-R has stronger multilingual generalization capacity and often performs better in cross-lingual and noisy text scenarios.

4.4.2 Training Setup

The model was fine-tuned using the following configuration:

Dataset split:

- 72% Training
- 8% Validation
- 20% Testing
- Learning rate: 2×10^{-5}
- Batch size: 16
- Number of epochs: 6
- Weight decay: 0.01

To address dataset imbalance—particularly the underrepresented "Threat" class—a class-weighted cross-entropy loss was applied. Class weights were computed using balanced class frequency estimation.

Model selection was based on validation performance, with macro F1-score used as a key evaluation metric due to class imbalance.

XLM-R demonstrated improved stability and better recognition of minority classes compared to the baseline BERT model.

4.5 Ensemble Model

To improve overall classification robustness and reduce prediction variance, an ensemble learning approach was implemented by combining the outputs of the fine-tuned BERT and XLM-R models. Instead of selecting a single best-performing transformer, probability-level fusion was applied to leverage strengths of both models.

The ensemble was constructed using soft voting on predicted class probabilities.

4.5.1 Soft Voting

Soft voting combines the predicted probability distributions from multiple models rather than their hard class labels. This approach preserves confidence information and generally provides more stable predictions.

- BERT produces a probability vector $\mathbf{P}_{\text{BERT}}$
- XLM-R produces a probability vector $\mathbf{P}_{\text{XLM-R}}$

Each vector contains probability scores for all five classes. Probabilities are aggregated and the final class is determined by the highest combined probability.

4.5.2 Probability Averaging

The ensemble probability distribution is computed as:

$$\mathbf{P}_{\text{ens}} = \frac{\mathbf{P}_{\text{BERT}} + \mathbf{P}_{\text{XLM-R}}}{2}. \tag{4.7}$$

where:

- $w_1 + w_2 = 1$
- w_1 is the weight assigned to BERT
- w_2 is the weight assigned to XLM-R

The final predicted class is determined as:

$$\hat{y} = \arg\max\left(\mathbf{P}_{\text{ens}}\right). \tag{4.8}$$

Weight selection was performed experimentally using validation performance. A grid search over different weight combinations was conducted to maximize overall accuracy and macro F1-score.

The ensemble model demonstrated improved generalization, particularly in under-represented categories such as "Threat," by combining contextual strengths of both transformer architectures.

Chapter 5

Results and Analysis

5.1 Classical Model Results

The classical model was evaluated using the two-stage hierarchical classification pipeline described in Chapter 4. Since the architecture first performs binary classification and then multi-class categorization, results are reported separately for each stage. Performance evaluation was conducted on the 20% test split using accuracy, precision, recall, F1-score, and confusion matrix analysis.

5.1.1 Stage-1: Binary Classification

In the first stage, the model distinguishes between:

- 0 → Not Bully
- 1 → Bully

Tag	Precision	Recall	F1-score	Support
Not Bully (0)	0.76	0.67	0.71	5321
Bully (1)	0.76	0.83	0.79	6710
Accuracy			0.76	12031
Macro avg	0.76	0.75	0.75	12031
Weighted avg	0.76	0.76	0.76	12031

Table 5.1: Binary Classification Performance

The binary classifier demonstrates moderate performance, effectively separating harmful from non-harmful comments.

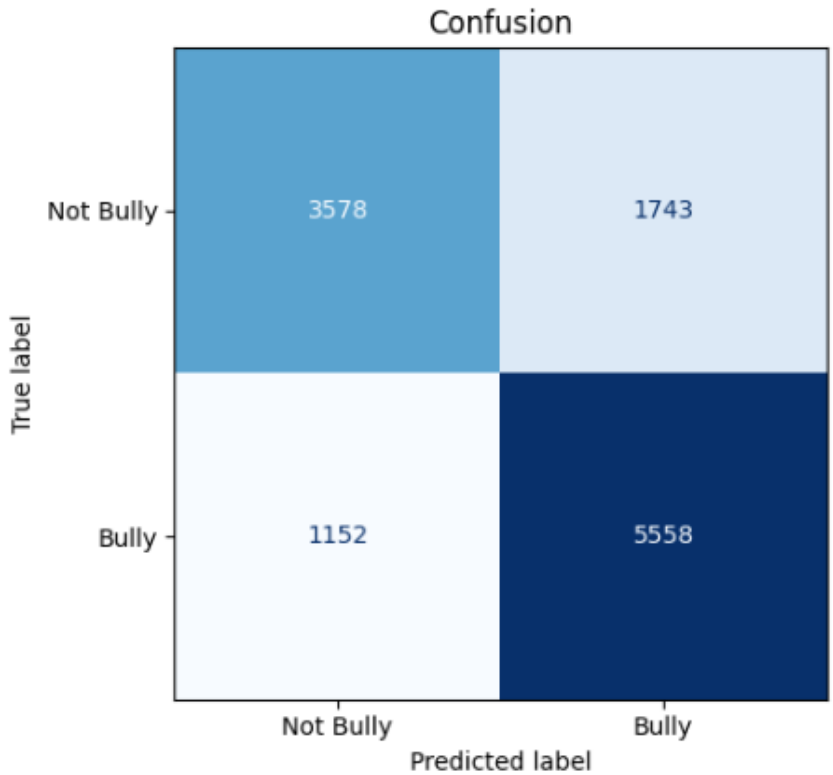

Figure 5.1: Confusion Matrix for Binary Classification

Analysis

- The model achieves higher recall for the Bully class (0.83) compared to the Not Bully class (0.67).
- False positives occur when aggressive but non-harmful language is misclassified as bullying.
- False negatives occur when subtle bullying expressions are misclassified as non-bullying.
- Overall, Stage-1 provides a stable filtering mechanism before category-level classification.

5.1.2 Stage-2: Multi-Class Classification

In the second stage, only comments predicted as "Bully" are categorized into specific bullying types.

Tag	Precision	Recall	F1-score	Support
Religious	0.56	0.59	0.57	1538
Sexual	0.57	0.49	0.53	2112
Threat	0.27	0.30	0.28	356
Troll	0.63	0.66	0.65	2704
Accuracy			0.57	6710
Macro avg	0.51	0.51	0.51	6710
Weighted avg	0.58	0.57	0.57	6710

Table 5.2: Multi-Class Performance

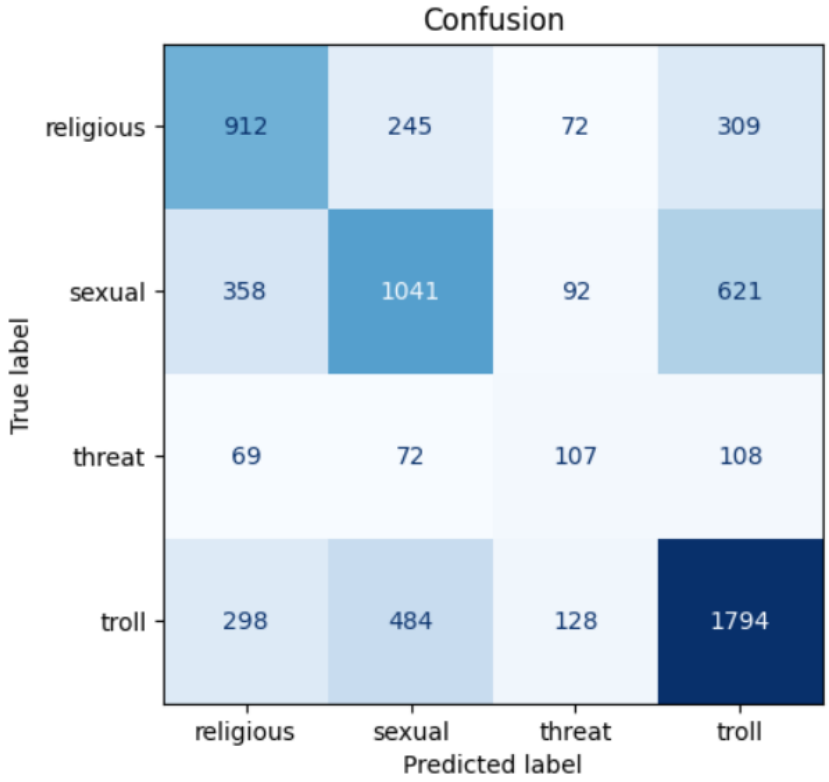

Figure 5.2: Confusion Matrix for Multi-Class Classification

Analysis

- The Troll category achieved the highest performance (F1 = 0.65), likely due to distinctive lexical patterns.
- The Threat category shows the lowest performance (F1 = 0.28), primarily due to limited sample size (2.96% of the dataset).
- Overlap between Religious and Troll categories indicates contextual similarity in aggressive phrasing.
- Performance drops significantly from Stage-1 to Stage-2, highlighting limitations of linear TF-IDF-based models in capturing fine-grained semantic differences.

5.2 BERT Results

Tag	Precision	Recall	F1-score	Support
0 (Not Bully)	0.8725	0.8400	0.8560	5401
1 (Threat)	0.6904	0.8145	0.7473	345
2 (Religious)	0.9227	0.9172	0.9199	1522
3 (Troll)	0.7169	0.7488	0.7325	2655
4 (Sexual)	0.8050	0.8164	0.8107	2119
Accuracy			0.8248	12042
Macro avg	0.8015	0.8274	0.8133	12042
Weighted avg	0.8274	0.8248	0.8257	12042

Table 5.3: BERT Performance

The fine-tuned Bangla BERT model was evaluated on the 20% test set using accuracy, precision, recall, macro F1-score, and confusion matrix analysis. Compared to the classical model, BERT demonstrated improved contextual understanding due to its bidirectional transformer architecture.

5.2.1 Training Curve

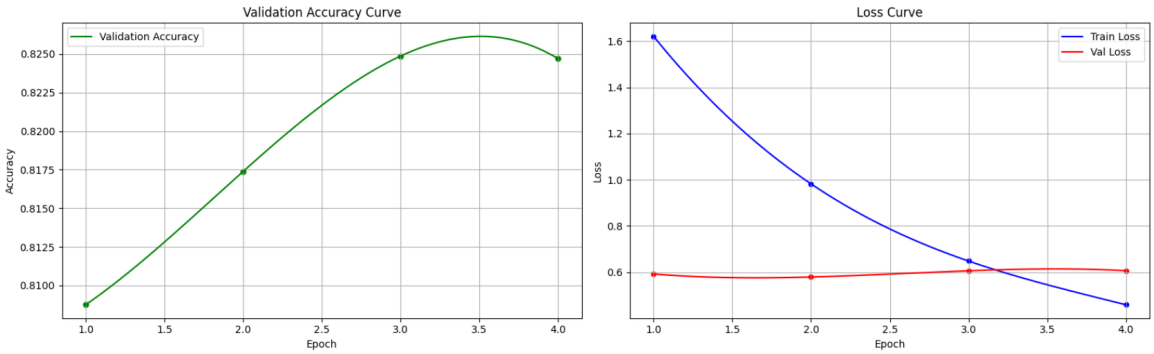

Figure 5.3: Accuracy & Loss Curve of BERT

During fine-tuning, both training and validation metrics were monitored across epochs. The training process showed:

- Gradual decrease in training loss
- Consistent reduction in validation loss
- Improvement in validation accuracy over initial epochs

The model converged within 4 epochs without severe overfitting. Early stopping was applied based on validation performance to prevent unnecessary training beyond optimal convergence.

The training curve indicates that BERT successfully adapted pretrained language representations to the cyberbullying classification task. Minor fluctuations in validation accuracy were observed, which is typical in transformer fine-tuning due to small batch sizes and dataset imbalance.

Overall, the model achieved stable convergence & strong generalization capability.

5.2.2 Confusion Matrix

The confusion matrix analysis reveals that BERT significantly improved category-level discrimination compared to the classical baseline.

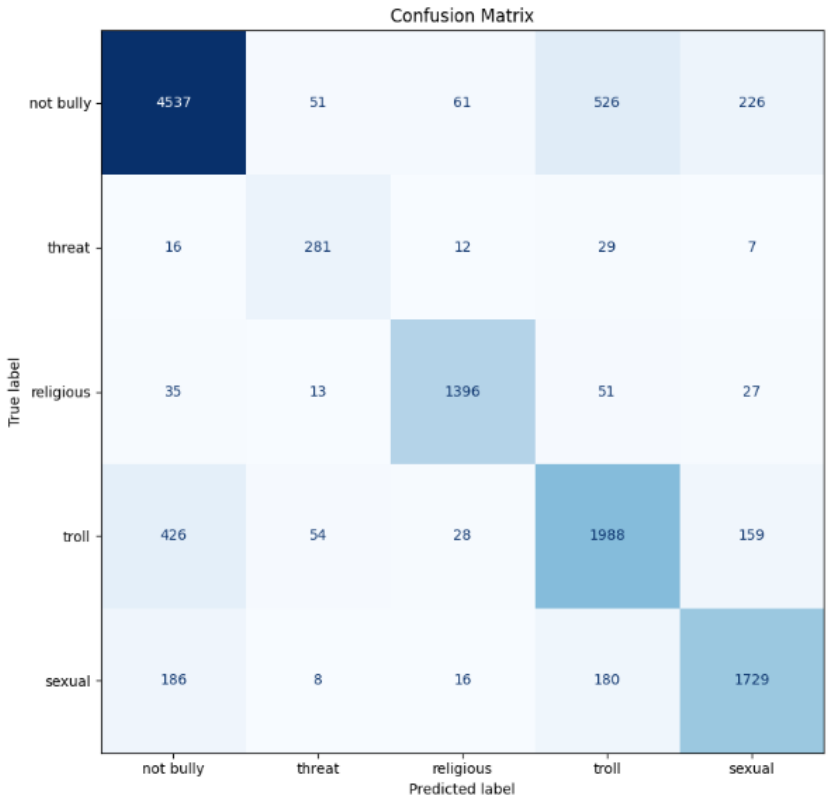

Figure 5.4: Confusion Matrix for BERT

Key observations include:

- High correct classification rate for "Not Bully" due to abundant training samples.
- Strong performance in identifying "Sexual" and "Troll" categories, which contain distinctive lexical patterns.
- Improved recognition of "Religious" compared to classical models.
- Some confusion between "Threat" and other bullying categories, likely due to limited sample size (2.96% of dataset).

BERT reduced false negatives in bullying detection, indicating better sensitivity to contextual aggression. However, minority class imbalance still affected performance for the "Threat" category.

Overall, BERT demonstrated superior contextual modeling ability and achieved higher macro F1-score than the classical approach.

5.3 XLM-R Results

Tag	Precision	Recall	F1-score	Support
0 (Not Bully)	0.8918	0.8473	0.8690	5401
1 (Threat)	0.7117	0.8087	0.7571	345
2 (Religious)	0.9088	0.9297	0.9191	1522
3 (Troll)	0.7527	0.7748	0.7635	2655
4 (Sexual)	0.8161	0.8584	0.8367	2119
Accuracy			0.8426	12042
Macro avg	0.8162	0.8438	0.8291	12042
Weighted avg	0.8448	0.8426	0.8432	12042

Table 5.4: XLM-R Performance

The fine-tuned XLM-RoBERTa (XLM-R) model was evaluated on the 20% test set using accuracy, precision, recall, and macro F1-score. Due to its multilingual pretraining and contextual representation strength, XLM-R demonstrated improved robustness, particularly for minority classes.

5.3.1 Training Curve

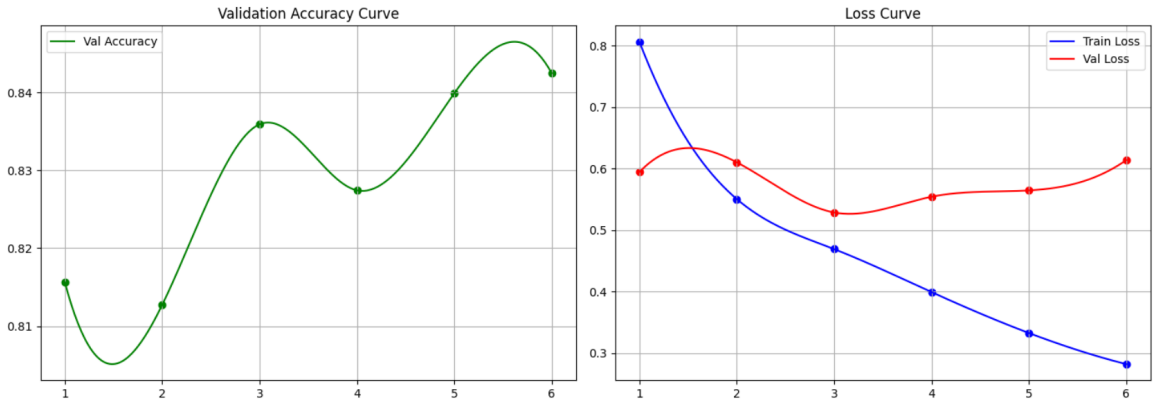

Figure 5.5: Accuracy & Loss Curve of XLM-R

During fine-tuning, training and validation metrics were monitored across 6 epochs. The training behavior showed:

- Steady decrease in training loss
- Gradual stabilization of validation loss
- Improvement in validation macro F1-score over early epochs

Unlike BERT, XLM-R was trained with a class-weighted loss function, which helped address dataset imbalance, especially for the underrepresented "Threat" class.

No significant overfitting was observed, as validation performance remained stable relative to training performance. The learning rate and weight decay settings contributed to smooth convergence.

Overall, the training curve indicates effective adaptation of pretrained multilingual representations to the Bangla cyberbullying classification task.

5.3.2 Confusion Matrix

The confusion matrix analysis highlights several improvements compared to both classical and BERT models.

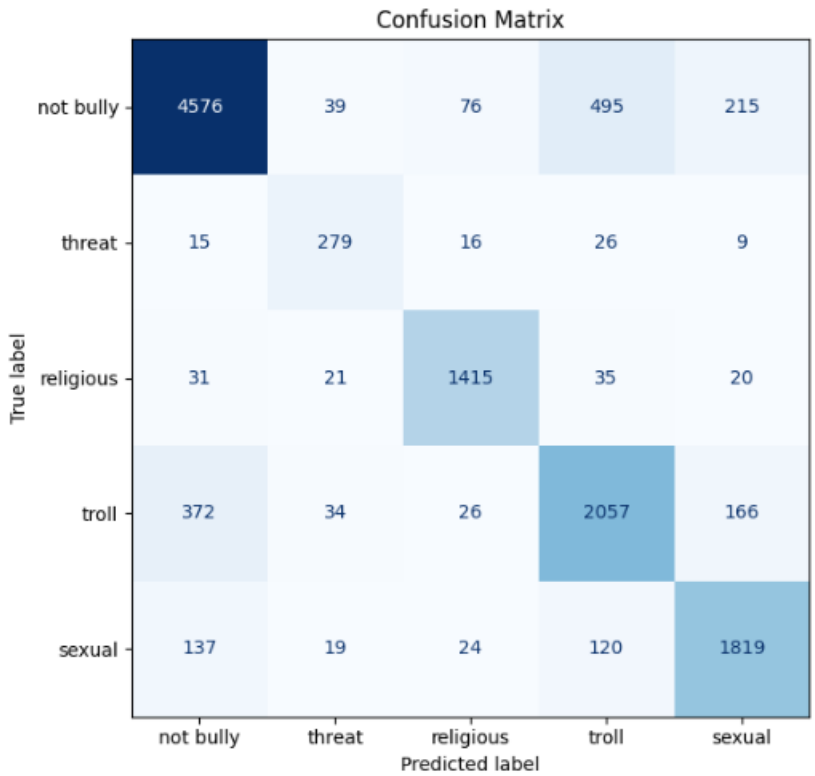

Figure 5.6: Confusion Matrix for XLM-R

Key observations:

- High correct classification rate for "Not Bully."
- Improved detection of "Threat" due to class-weighted loss.
- Better separation between "Religious" and "Troll" categories compared to classical models.
- Reduced misclassification among semantically similar bullying types.

Although minor confusion between certain categories still exists, XLM-R achieved better macro-level balance across classes, particularly for minority categories.

Overall, XLM-R demonstrated stronger generalization and improved sensitivity to subtle contextual differences compared to BERT and classical models.

5.4 Ensemble Results

To improve classification robustness and reduce prediction variance, a soft-voting ensemble approach was implemented by combining the probability outputs of BERT and XLM-R models. Instead of relying on a single transformer architecture, probability-level fusion was applied to leverage complementary contextual strengths.

The optimal weight configuration obtained through validation tuning was:

$$w_{\text{bert}} = 0.475, \quad w_{\text{xlmr}} = 0.525$$

Tag	Precision	Recall	F1-score	Support
Not Bully	0.9519	0.9188	0.9351	5173
Threat	0.8329	0.9598	0.8919	348
Religious	0.9609	0.9602	0.9605	1483
Troll	0.8673	0.8932	0.8801	2707
Sexual	0.9128	0.9340	0.9233	2029
Accuracy			0.9220	11740
Macro avg	0.9052	0.9332	0.9182	11740
Weighted avg	0.9233	0.9220	0.9223	11740

Table 5.5: Ensemble Performance

The ensemble model demonstrated the highest overall performance among all implemented approaches.

5.4.1 Confusion Matrix Analysis

To further analyze classification behavior, the confusion matrix of the ensemble model was examined.

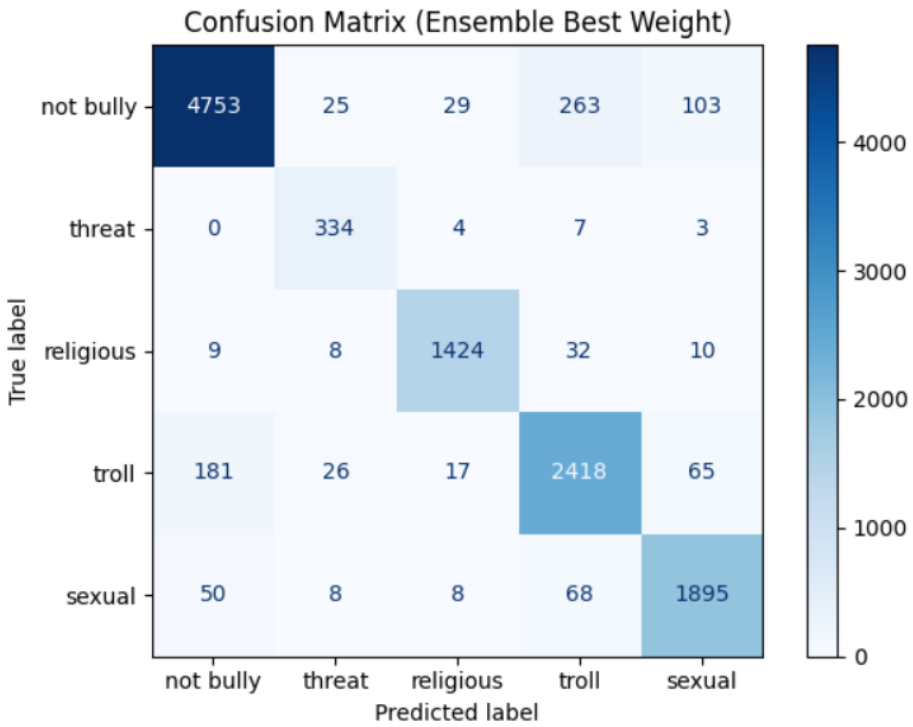

Figure 5.7: Confusion Matrix for Ensemble Model

The confusion matrix indicates:

- High true positive rates for Not Bully and Troll categories.
- Improved correct prediction for the Threat class compared to individual transformer models.
- Reduced cross-category confusion between Religious and Troll.
- Balanced distribution of predictions across all five classes.

Unlike the classical model, the ensemble demonstrates stronger diagonal dominance in the confusion matrix, indicating more accurate class-specific predictions.

5.4.2 Real-Time Demonstration

To validate practical applicability, the trained ensemble model was deployed using a lightweight real-time interface (Streamlit-based application). The system allows users to input Bangla text and receive immediate classification output.

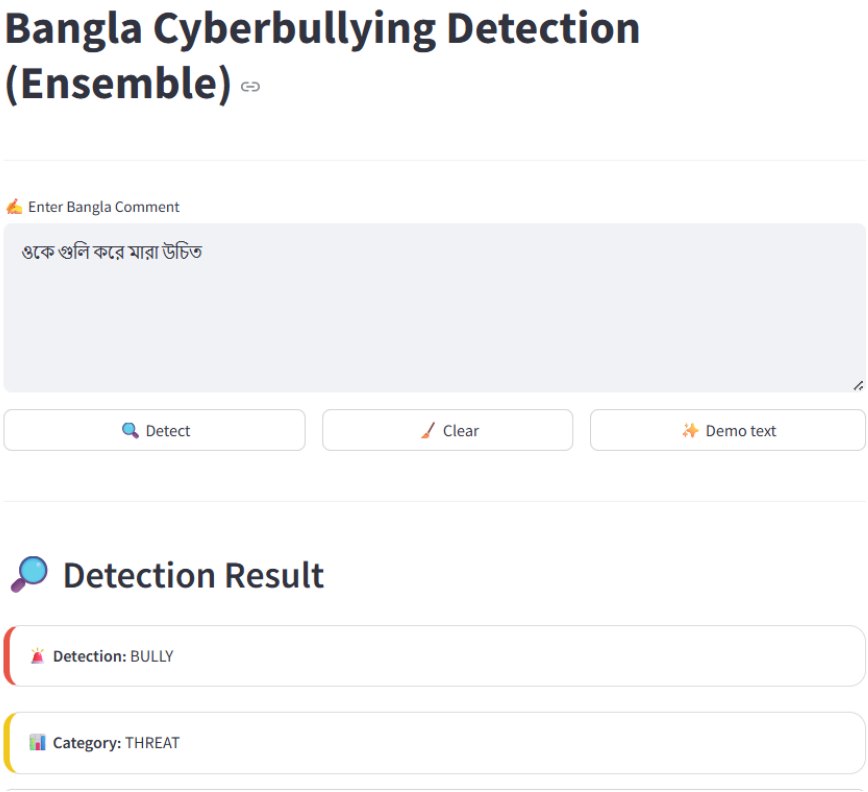

Figure 5.8: Real-time interface on Streamlit

System Workflow in Real-Time: The user inputs a Bangla comment, which is tokenized using a pretrained tokenizer. BERT and XLM-R generate probability distributions, and a soft-voting ensemble combines them to compute the final probability. The class with the highest probability is returned as the final prediction.

Demonstration Observations: The model responds instantly with stable and consistent predictions. Confidence scores help interpret prediction reliability, and the system performs well on unseen user-generated text. Short and direct abusive comments are classified with high confidence.

The real-time implementation confirms that the ensemble model is not only experimentally strong but also practically deployable for moderation systems.

5.5 Model Performance Comparison

Model	Accuracy	Precision	Recall	Macro F1-score
Classical (Binary Stage)	75.94%	75.89%	75.04%	75.27%
Classical (4-Class Stage)	57.44%	50.60%	51.25%	50.82%
BERT	82.47%	82.74%	82.47%	81.32%
XLM-R	84.25%	84.47%	84.25%	84.31%
Ensemble	92.20%	90.52%	93.32%	91.82%

Table 5.6: Performance Comparison Table

Observations

- Transformer-based models significantly outperform classical approaches.
- XLM-R performs better than BERT in all macro-level metrics.
- The ensemble improves accuracy by:
  - +7.95% over XLM-R
  - +9.73% over BERT
- Macro F1-score increases substantially, indicating better class balance handling.

The ensemble achieves the highest and most stable performance across all evaluation metrics.

5.5.1 Error Analysis

A detailed examination of misclassified samples reveals several important patterns.

1. Minority Class

The "Threat" class constitutes only 2.96% of the dataset. In individual models, this category suffered from lower recall due to limited training samples. However, the ensemble model significantly improved recall, reducing false negatives.

2. Category Overlap
Confusion still occurs between:

- Troll and Religious
- Sexual and Troll

This is primarily due to overlapping linguistic patterns and aggressive tone similarities in Bangla social media text.

3. Implicit and Sarcastic Expressions
Subtle sarcasm and indirect aggression remain challenging. Even transformer models occasionally misclassify such comments due to contextual ambiguity.

4. Short Text Limitation
Very short comments lack sufficient semantic context, increasing classification difficulty across all models.

Chapter 6

Conclusion

6.1 Summary of Findings

This research focused on developing an automated system for detecting cyberbullying in Bangla text using machine learning and deep learning approaches. The primary objective of the study was to classify Bangla social media comments into bullying and non-bullying categories with acceptable performance. A large-scale dataset containing thousands of Bangla text samples was used to train and evaluate the models. Before model training, necessary preprocessing steps such as removing special characters, punctuation, extra spaces, and stopwords were applied to improve data quality. The cleaned text data was then transformed into numerical representations so that classification algorithms could process it effectively.

Several classical machine learning models as well as advanced transformer-based deep learning models were implemented and compared. The experimental results demonstrated that while traditional models were capable of identifying explicit offensive language, deep learning and transformer-based models showed better contextual understanding of Bangla text. These models achieved improved performance in terms of accuracy, precision, recall, and F1-score. Furthermore, the ensemble approach contributed to more stable and reliable predictions by combining the strengths of multiple models. Overall, the findings of this study indicate that automated detection of Bangla cyberbullying is both feasible and effective when appropriate preprocessing techniques and modern NLP models are applied.

6.2 Limitations

Despite achieving promising results, this study has several limitations. Although a relatively large dataset was used, it may not fully represent the diversity of Bangla language expressions found across different regions and online communities. Bangla social media text often includes slang, informal spelling, and mixed-language writing (Banglish), which increases the complexity of classification tasks. In addition, detecting sarcasm, indirect harassment, and context-dependent bullying remains challenging for automated systems. Some misclassifications occurred when harmful intent was expressed without explicit abusive words or when offensive words were used in non-harmful contexts.

Another limitation is related to computational resources. Since the models were trained using Google Colab, there were restrictions in terms of processing power and training time, which limited extensive hyperparameter tuning and experimentation with larger transformer architectures. Moreover, the system was evaluated in an experimental setup and was not deployed in a real-time environment. Therefore, its real-world performance may vary when integrated into live social media platforms.

6.3 Future Work

Future research can focus on expanding the dataset by collecting more diverse and real-world Bangla comments from various online platforms. A larger and more balanced dataset would help improve model generalization and robustness. Advanced Bangla-specific pretrained transformer models can be further fine-tuned to enhance contextual understanding and semantic interpretation. Instead of limiting the system to binary classification, future studies can implement multi-class classification to identify different types of cyberbullying, such as hate speech, threats, harassment, and offensive language.

Additionally, integrating sentiment analysis and emotion detection techniques may provide deeper insight into harmful online behavior. Developing a real-time web-based or mobile-based application for automatic detection and filtering of cyberbullying content would significantly increase the practical value of this research. Finally, future work can explore cross-lingual and multilingual models to better handle mixed Bangla-English text commonly found on social media platforms.

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